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Implementation and Evaluation of Shocks and Vibrations Dampening Tools in Challenging Unconventional Drilling Applications

Hassan Almomen, Saudi Aramco; Anton Shatrovskiy and Nikita Dremov, SLB

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Abstract

Shock and vibration while drilling vertical open hole section through various formations layers are the main challenges that increase risk of bit damage and downhole tool failure. The most common method of mitigating shock and vibration is controlling drilling parameters which can impact well delivery objectives. Shock and vibration dampening tools were introduced and evaluated against benchmarks in the field as an innovative solution that does not require controlling drilling parameters and performance.

Three different types of shock and vibration dampening tools from different providers with different technology were tested. Their placement in the drill string varied in between Rotary Steerable System (RSS) and Measurement While Drilling Tool (MWD). Analysis was done by comparing results in the same field with identical wellbore profile and using the appropriate shock and vibration measuring sensors to acquire the most accurate analysis. Three axis shock and vibration sensors were installed on both RSS and MWD tools.

Initial observations revealed marginal shock and vibration level reduction on some of prior runs, however it was observed that all these runs were having shock and vibration dampening tools placed above Bottom hole assembly (BHA). This placement does not take advantage of the unique wireless communication feature with RSS system allowing for extra tools such as shock and vibration dampening tools, measurement sensors and Reamers being placed close to the bit between RSS and MWD. It is recommended to place shock and vibration dampening tools between RSS and MWD for optimum benefits. This innovative idea reduced shock and vibration by 50 percent on MWD tool and increased vertical ROP by 35 percent making a major step change in drilling performance. This initiative saved several operational days when considering multiple wells drilled per year. Furthermore, this initiative provided a secondary benefit of reducing repair, maintenance time and manpower required to fix damage associated with excessive shock and vibration.

This paper illustrates the benefits of utilizing dampening tools to overcome the shock and vibration, analyze placement in the BHA and discuss conclusions on added value to deliver the optimum performance.

Introduction

Drilling in unconventional field faced numerous technical challenges, particularly in managing the temperature profile, drilling dynamics, and shocks and vibration (S&V) management. Bottom-hole assemblies were configured to be compatible with optimally selected drill bits to thrive through the harsh drilling dynamics. Real-time monitoring systems were implemented to track drilling parameters and adjust operations dynamically.

Managing shocks and vibrations while drilling the vertical part of the section and traversing various formation layers and transitions is a significant challenge. These issues pose a high risk of drill string and bit damage, as well as downhole tool failures. The most common method of mitigating shocks and vibrations is by controlling drilling parameters. However, this approach comes with a high cost to drilling performance and well delivery objective which driven to find innovative solutions to shocks and vibrations that do not involve controlling parameters and performance. To address this, shocks and vibrations dampening tools were introduced and evaluated against field benchmarks. Three different types of shocks and vibrations dampening tools from various providers, each utilizing different technologies, were tested. Their placement in the drill string varied between the Rotary Steerable System (RSS) and the Measurement While Drilling Tool (MWD). The analysis was conducted by comparing results in the same field with a typical wellbore profile, using appropriate measuring sensors to acquire the most accurate analysis. Three-axis S&V sensors were installed on both RSS and MWD tools.

Initial studies revealed marginal S&V level reduction in some prior runs. However, it was observed that these runs had S&V dampening tools placed above the Bottom-Hole Assembly (BHA), which did not take full advantage of the wireless communication feature of RSS system. This feature allows for the insertion of additional tools (S&V dampening tools, measurement sensors, reamers, etc.) close to the bit, between the RSS and MWD.

Vibrations Forms

Drill string S&V is divided into three basic forms such as axial (longitudinal) mode, torsional (rotational) mode and transverse (lateral) mode. When the drill string moves along its axis of rotation, this is called axial vibration/shock. Torsional vibrations are caused due to an irregular rotation of the drill string when rotated from the surface at constant speed. When the drill string moves laterally to its axis of rotation it is called lateral S&V. Due to many factors, very complex drill string S&V are produced during actual drilling. According to past theories, indoor experiments, and field studies, the relationship of S&V is listed, which contains basic forms, modes, frequencies, amplitudes response and tool damage.

Axial vibrations

Axial vibrations of a drill string have been well studied and documented throughout the years. This mode of vibration generates bouncing in the direction along the axis of the drill string in the wellbore direction as shown in Figure 2. Axial vibrations are caused by the movement of the drill string, upwards and downwards, and may induce bit bounce. Axial vibrations are damped by the drill string itself due to the stiffness in the length direction and can be directly detected by the driller at shallow depths, as the vibrations travels to surface through the drill string. This mode of vibration is considered less aggressive than the other modes and the recorded axial accelerations are usually significantly lower, due to the large masses that have to be set in motion.

Real-time mitigation actions include adjusting the revolutions per minute (RPM) and weight on bit (WOB), by increasing the WOB and reducing the RPM, to change the drill string energy. If this does not work, it is recommended to stop drilling to allow the vibrations to cease and thereafter start drilling with different parameters. This must be done in correlation with the ROP, as WOB and RPM are the most highlighted parameters affecting the drilling speed.

Lateral / Transverse vibrations

Lateral vibrations are seen as side-to-side motion in transverse direction relative to the string, illustrated in Figure 1. The vibration mode is primarily generated by whirl. Whirl is the eccentric rotation of the drill string, or part of it, around a point other than the geometric center of the borehole. This motion will only occur if there is enough lateral movement in the BHA to bend out and touch the borehole wall. In severe cases it is known for triggering both axial and torsional vibrations, a phenomenon called mode coupling.

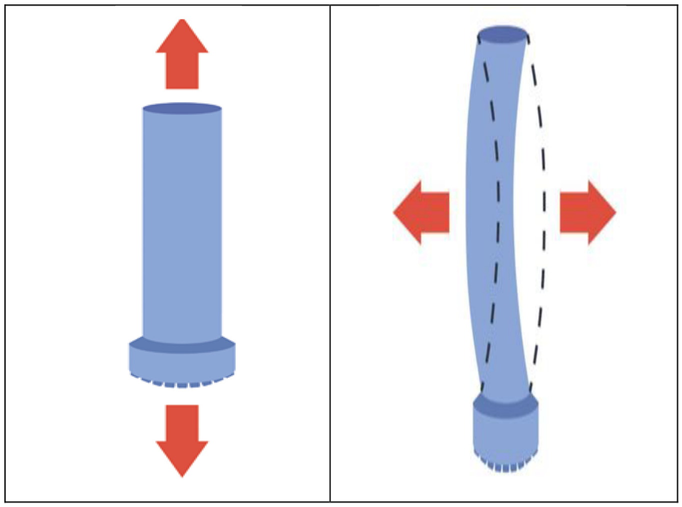

Figure 1—Axial and Lateral Vibrations

Transverse vibrations are viewed by the industry as the most destructive mode, causing severe damage to the BHA components and wellbore, as the bit and BHA continuously impact the wellbore. The interaction between the bit and BHA with the wellbore wall leads to operational troubles such as overgauge holes, damaged equipment, lack of well direction control and drill string fatigue.

Evaluation Technology

The practical measuring method is another effective evaluation method of drill string. In order to obtain dynamic failure model range and experience model, the measured S&V data and the observed failure are combined. Hence, the vibration measurement tools are the key technology to determine the accuracy and stability of evaluation approach.

The applicable scope of practical measuring method to evaluate the drill string S&V levels is limited by the method accuracy, because different areas have different stratigraphic characteristics. The practical measuring method must be based on several measures and analyses. This is the developed S&V Limit based on huge experience and post analysis on damage's tools. Table 1 shows the level of sever the S&V is classified based on ranges agreed. In drilling operation, once vibration readings reach the limits, this will directly raise the alert to control/adjust the drilling parameters to avoid tool failure which will slow down the operation and cause extra trip out of hole to change the tool. Reaching the red limit indicates that WOB should be reduced gradually until shocks level decrease. However, reaching the black limit indicated the need to stop drilling, pick-off bottom and reset parameters before resuming drilling again.

Table 1—S&V Limits

MWD (G measurements)			RSS (G measurements)			
VIB_LAT	VIB_AXL	SHKPK_LAT	VIB_LAT	VIB_AXL	SHKPK_LAT	Action
< 2	< 4	< 12	< 16	< 3	< 110	Low Risk
2 - 4	8 - 12	16 - 20	16 - 18	3 – 5	< 160	Working Range for Performance
4 - 7	16	24	18 - 24	5 - 7	< 225	Must mitigate - Reduce WOB by 2Klbs WOB
> 7	> 16	> 24	> 24	> 7	> 225	STOP, pick up off bottom, reset parameters

Methodology

The main objective is to provide an insight into the primary vibration problem and determine effective mitigation tools and techniques, particularly concerning the Bottom-Hole Assembly (BHA) design, which can minimize shock and vibration and improve drilling performance in the future. A particular focus is given to dampening tools and their placement within the BHA.

Dampening tools are essential in drilling operations to reduce vibrations and shocks, ensuring smoother and more efficient drilling. Table 2 shows dampening tool components, specifications, limitations and general application:

Table 2—Dampening tool components, specifications, limitations and general application

	Description
Internal Components	• Springs: Often steel disk springs to absorb shocks. • Hydraulic Mechanisms: Used to control and dampen vibrations. • Seals: Metal-to-metal seals for durability in harsh conditions.
Specifications	• Size: Varies widely, typically ranging from 1-11/16″ to 12-1/4″ in diameter. • Material: High-strength steel and nonmagnetic alloys. • Operating Conditions: Can handle high temperatures (up to 500°F) and aggressive drilling fluids. • Spring Rates: Adjustable to suit different drilling conditions.
Limitations	• Complexity: Requires precise control and monitoring. • Maintenance: Regular maintenance needed to ensure optimal performance. • Cost: Can be expensive due to advanced materials and technology.
Applications	• Vibration Reduction: Minimizes bit bounce and impact loads. • Tool Protection: Extends the life of bits, motors, RSS and surface equipment. • Efficiency: Enhances drilling efficiency and reduces costs. • High-Vibration Environments: Ideal for challenging formations. • Geothermal Drilling: Reliable performance in high-temperature environments.

The strategic placement of shock and vibration (S&V) dampening tools within the Bottom-Hole Assembly (BHA) enables a comprehensive analysis of S&V reduction facilitated by advanced smart tools equipped with sensors. These sensors, integrated within the Measurement While Drilling (MWD) and Rotary Steerable System (RSS), are capable of detecting various types of vibrations throughout the BHA. The placement of shock and vibration (S&V) dampening tools within the Bottom-Hole Assembly (BHA) can be optimized in two different configurations: one option is to place the dampening tool on top of the BHA, and the second option is to position it between the Rotary Steerable System (RSS) and the Measurement While Drilling (MWD) tool, as illustrated in Figure 2.

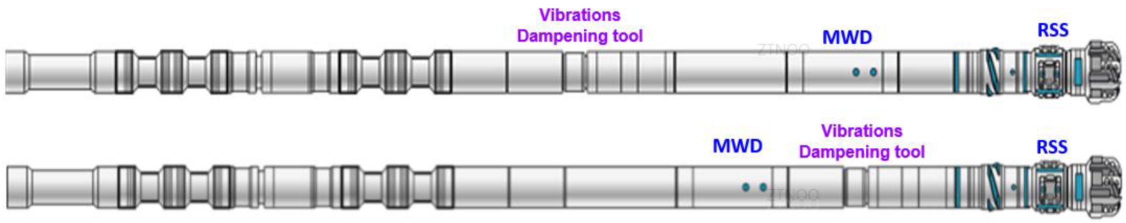

Figure 2—Different Placement of the Vibration dampening tool in the BHA

Results and Discussion

Vibrations on the MWD tool

Comparative analysis was conducted to evaluate the lateral vibrations on the MWD tool, as illustrated in Figure 3. First row without dampening tool, second row with dampening tool above MWD, the third row with dampening tool between MWD and RSS for 3 different wells with same well profile and BHA, within the same field.

The figure 3 below illustrates the distribution of shock levels experienced during drilling operations, categorized into four distinct levels: low (green), moderate (yellow), high (red), and severe (black). The x-axis represents "Surface WOB, klbf" ranging from 0 to 60, while the y-axis represents "Lateral Vibrations, GRMS" ranging from 0 to 50.

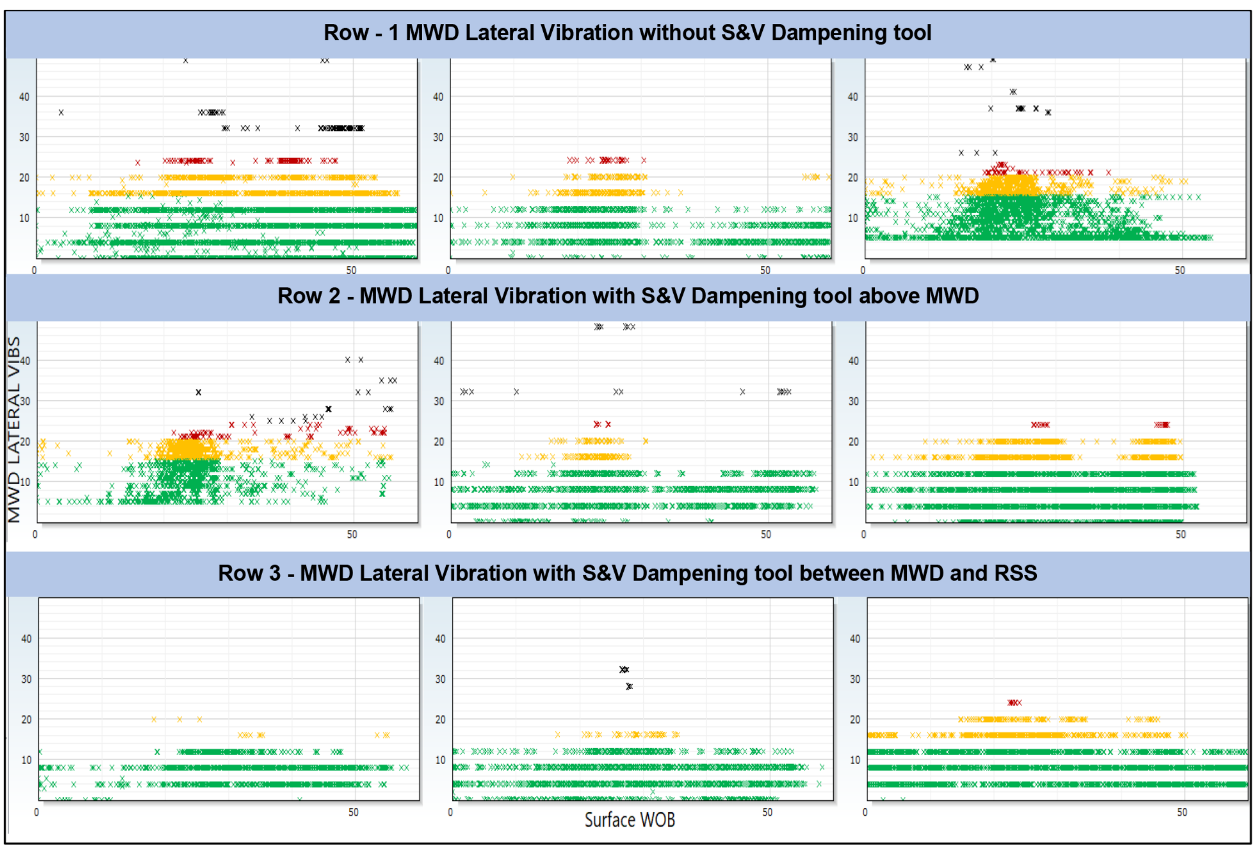

Figure 3—Offsets wells analysis for MWD Lateral vibrations

In the first row, the scatter plots show the relationship between Surface WOB and MWD Lateral Vibrations for three wells without any dampening tools. The data points indicate that as WOB increases, lateral vibrations also increase significantly. Green dots dominate at lower WOB values (0-20) klbf, but yellow, red, and black dots become more frequent as WOB rises, indicating moderate to severe vibrations. This suggests that without dampening tools, the BHA experiences high levels of lateral vibrations, which can compromise its integrity and performance.

The second row represents three wells with a dampening tool placed above the MWD. The scatter plots show a noticeable reduction in lateral vibrations compared to the first row. Green dots are more prevalent across a wider range of WOB values (0-30) klbf, indicating lower vibrations. Yellow and red dots appear less frequently, and black dots are less and scattered, even at higher WOB values (30-50) klbf. This configuration helps mitigate vibrations, enhancing the stability and performance of the BHA.

In the third row, the scatter plots illustrate the relationship between Surface WOB and MWD Lateral Vibrations for three wells with a dampening tool placed between the Rotary Steerable System (RSS) and MWD. The data points show a further reduction in lateral vibrations compared to the second row. Green dots dominate across almost all WOB values (0-40) klbf, with fewer yellow and orange dots. Black dots are rare, indicating minimal severe vibrations. This configuration provides optimal vibration dampening, ensuring better BHA integrity and drilling efficiency.

Overall, the use of dampening tools, especially when placed between RSS and MWD, significantly reduces lateral vibrations, improving the stability and performance of the BHA in drilling operations.

Vibrations on the RSS tool

A comparative analysis was conducted to evaluate the lateral vibrations on the RSS tool with various placements of the dampening tool. The results are illustrated below, Fig.4.

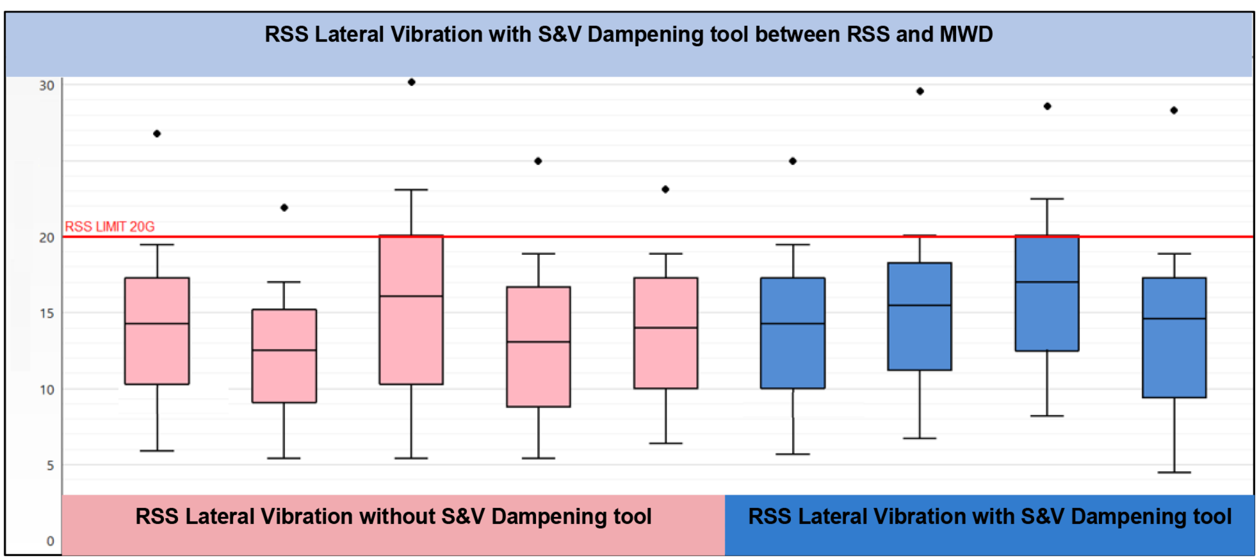

Figure 4—Offsets wells analysis for RSS Lateral vibrations

The box plot consists of nine box-and-whisker plots, each representing one well. The pink box plots show RSS Lateral Vibrations without the S&V Dampening tool, while the blue box plots represent RSS Lateral Vibrations with the dampening tool. The y-axis ranges from 0 to 30, with a red horizontal line labeled "RSS LIMIT 20G" at the y-value of 20 GRMS.

The pink box plots indicate the lateral vibrations experienced in wells without the use of the S&V Dampening tool. The medians of these plots range from 14 to 18, with interquartile ranges (IQRs) showing variability in the data. Several outliers exceed the RSS limit of 20G, indicating instances of severe vibrations.

This suggests that without the dampening tool, the wells experience higher levels of lateral vibrations, which can compromise the integrity and performance of the BHA.

The blue box plots represent the lateral vibrations in wells with the S&V Dampening tool. The medians of these plots range from 13 to 16, showing a slight reduction in vibration levels compared to the pink plots. The IQRs are similar, but there are fewer outliers exceeding the RSS limit of 20G. This indicates that the use of the dampening tool helps mitigate lateral vibrations, reducing the instances of severe vibrations and enhancing the stability and performance of the BHA.

The comparison between the pink and blue box plots highlights the effectiveness of the S&V Dampening tool in reducing lateral vibrations. While both sets of plots show variability, the blue plots have lower median values and fewer outliers exceeding the RSS limit of 20G. This suggests that the dampening tool is beneficial in maintaining BHA integrity and optimizing drilling performance.

Optimized BHA Design & Drilling ROP

The placement of S&V Dampening Tools between the Rotary Steerable System (RSS) and Measurement While Drilling (MWD) tools has proven to be highly effective. This innovative approach has resulted in a significant reduction of shock and vibration (S&V) levels by approximately 50% on the MWD tool. Consequently, the rate of penetration (ROP) has increased by 35%, marking a substantial improvement in drilling performance. This enhancement translates to a savings considering the rig rate and numbers of wells drilled. Additionally, there are considerable hidden savings in repair and maintenance costs, as well as reduced manpower requirements for addressing damages associated with excessive S&V.

The figure 5 below shows two plots representing the drilling mechanic log for two wells. Each plot contains sections on Y-axis labeled "ROP" and "Lateral MWD Vibrations" with various colored segments indicating different levels of vibrations. X-axis represent measured depth.

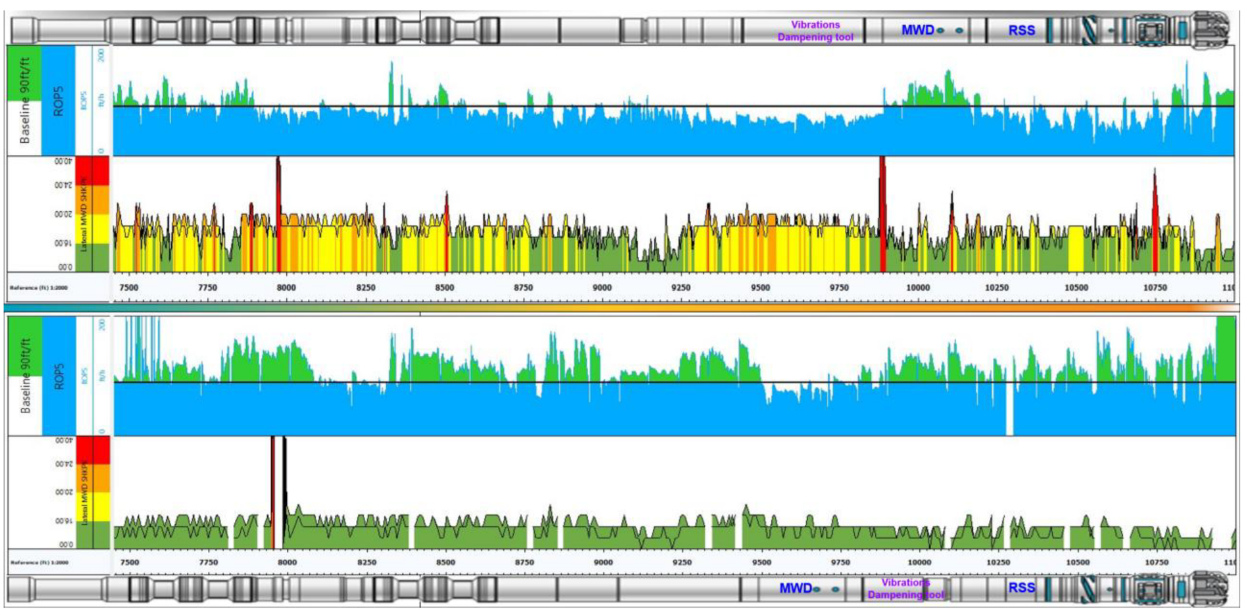

Figure 5—Effect of different placement S&V dampener for MWD Lateral vibrations

The top plot represents the drilling mechanic log for the first well, where the dampening tool is placed above the Measurement While Drilling (MWD) tool. The "Vibrations" section shows a range of colors from green to red, indicating varying levels of vibrations recorded by the MWD tool throughout the drilling process.

The bottom plot represents the drilling mechanic log for the second well, where the dampening tool is placed between the Rotary Steerable System (RSS) and MWD. The "Vibrations" section in this plot shows a more consistent presence of green segments. This indicates that placing the dampening tool between the RSS and MWD provides better vibration control, resulting in lower overall vibration levels recorded by the MWD tool. The reduced presence of red segments suggests improved BHA stability and performance.

In addition to the improved vibration control, the second well also shows a notable improvement in the Rate of Penetration (ROP). The optimized placement of the dampening tool between the RSS and MWD not only reduces vibrations but also enhances drilling efficiency. This configuration allows for smoother drilling operations, leading to a higher ROP compared to the first well. The increased ROP indicates more efficient drilling, reducing the overall time and cost associated with the drilling process.

Comparing the two plots, it is evident that the configuration of the dampening tool significantly affects the vibration levels recorded by the MWD tool during drilling. The second well, with the dampening tool between RSS and MWD, shows more effective vibration reduction and a notable improvement in ROP. This highlights the importance of optimal placement of dampening tools in drilling operations to achieve better performance, maintain BHA integrity, and enhance drilling efficiency.

Conclusion & Recommendations

- The innovative methodology of adding dampening tool in drilling BHA has resulted in a significant reduction of shock and vibration (S&V) levels by approximately 50% on the MWD tool.
- Improved overall ROP in 8-1/2" Section vertical part of 35% compared to runs without shock dampener tool.
- The placement of dampening tool between RSS and MWD shows more effective in vibration reduction and a notable improvement in ROP. This highlights the importance of optimal placement of dampening tools in drilling operations to achieve better performance, maintain BHA integrity, and enhance drilling efficiency.
- Overall, the use of dampening tools, especially when placed between RSS and MWD, significantly reduces lateral vibrations, improving the stability and performance of the BHA in drilling operations.
- The configuration of the dampening tool significantly affects the vibration levels recorded by the MWD tool during drilling. The second well, with the dampening tool between RSS and MWD, shows more effective vibration reduction and a notable improvement in ROP. This highlights the importance of optimal placement of dampening tools in drilling operations to achieve better performance, maintain BHA integrity, and enhance drilling efficiency.

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